Machine Learning in NBA Over/Under Betting

Business 35137: Machine Learning in Finance
The University of Chicago Booth School of Business

Alan Donnelly | Andrew McLaughlin | Robert Asgeirsson | Chris Mulligan

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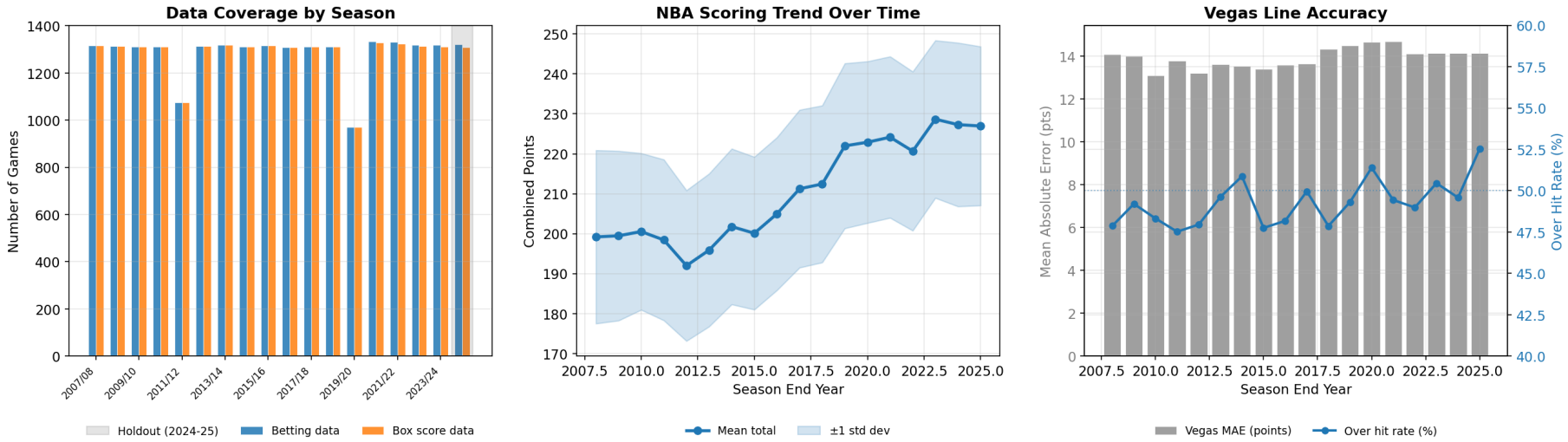
Research Question

- Can machine learning models predict NBA game totals more accurately than the Vegas line — and profitably?
- Standard −110 odds require winning 52.4% of bets to break even
- The Vegas line is the most efficient public forecast; any exploitable edge must be small and repeatable
- Approach: 6 model classes, walk-forward validation on 11 NBA seasons (2013/14 – 2023/24), true holdout test on 2024/25

Data and Feature Engineering

- 23,000 games across 18 NBA seasons (2007/08 – 2024/25); merged betting lines + box-score statistics
- 114 features in 7 categories: Vegas line, scoring, shooting, rebounding, playmaking, schedule, interactions
- Strictly lagged rolling windows (L5, L10) to prevent lookahead bias; 2024/25 season held out entirely

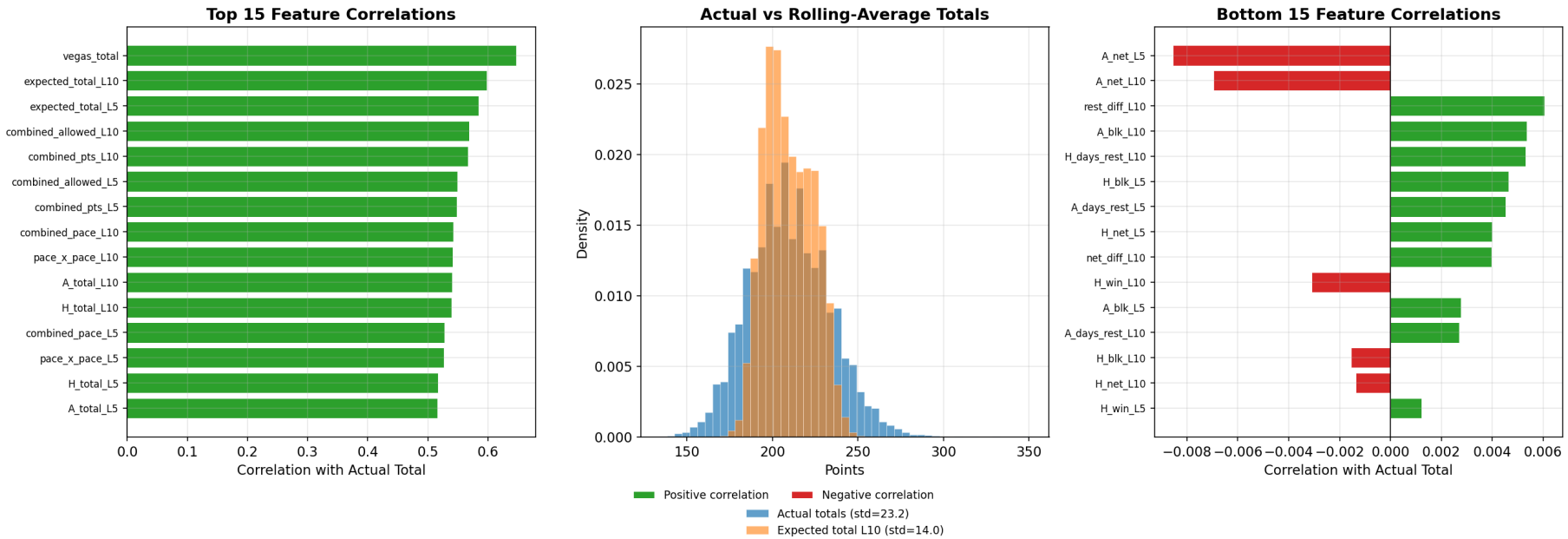
Figure 1: Data Overview — 2007-08 through 2024-25 NBA Seasons



Feature Signal Analysis

- Vegas-derived features dominate (r = 0.55–0.65); schedule/rest features show near-zero marginal correlation
- Interaction features (pace × pace, spread × total) help linear models capture non-additive effects

Figure 3: Feature Signal Preview — Correlation with Actual Total



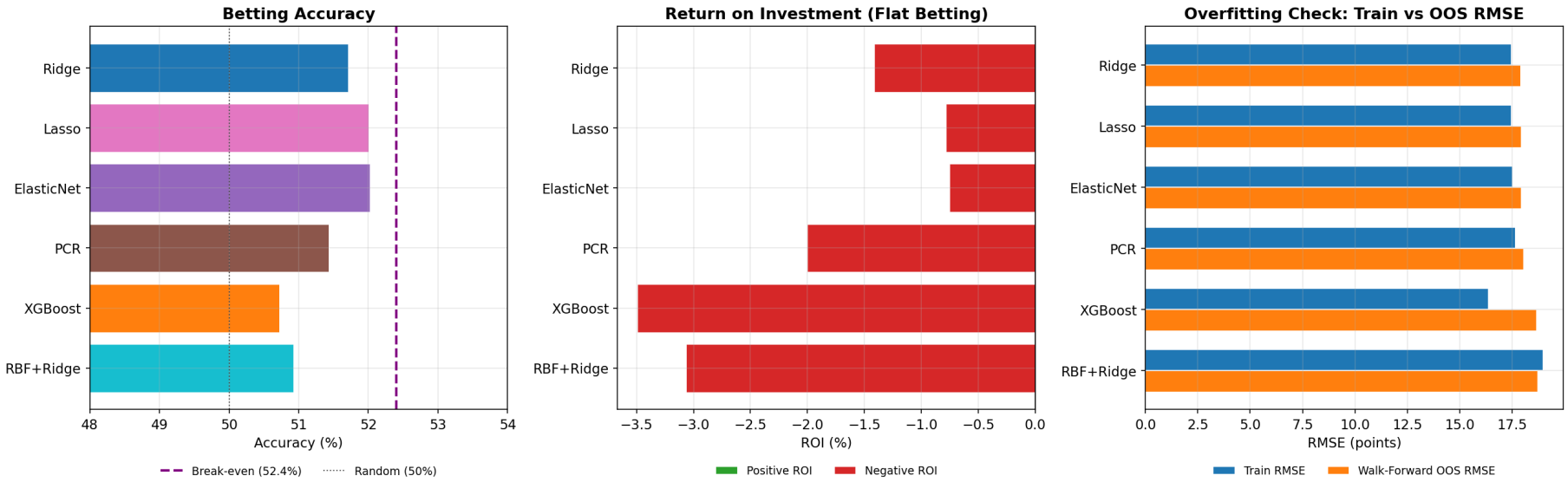
Walk-Forward Backtesting

- Strict temporal ordering: train on ALL prior data, predict one season ahead → 11 test seasons (2013/14 – 2023/24)
- Training set grows by ~1,300 games per iteration
- Hyperparameters tuned via 5-fold TimeSeriesSplit, optimising betting accuracy (not RMSE)
- Exponential recency weighting with tuned half-life — adapts to the NBA's evolving pace and scoring
- 6 models: Ridge, Lasso, ElasticNet, PCR, XGBoost, RBF+Ridge

Model Comparison: Simple Beats Complex

- Linear models (51.7–52.0%) outperform complex models (50.7–50.9%); no model reaches 52.4% break-even with flat betting
- XGBoost: 2.3-point train-vs-OOS RMSE gap → overfitting confirmed
- Bias-variance tradeoff: flexibility is a liability in low-signal environments

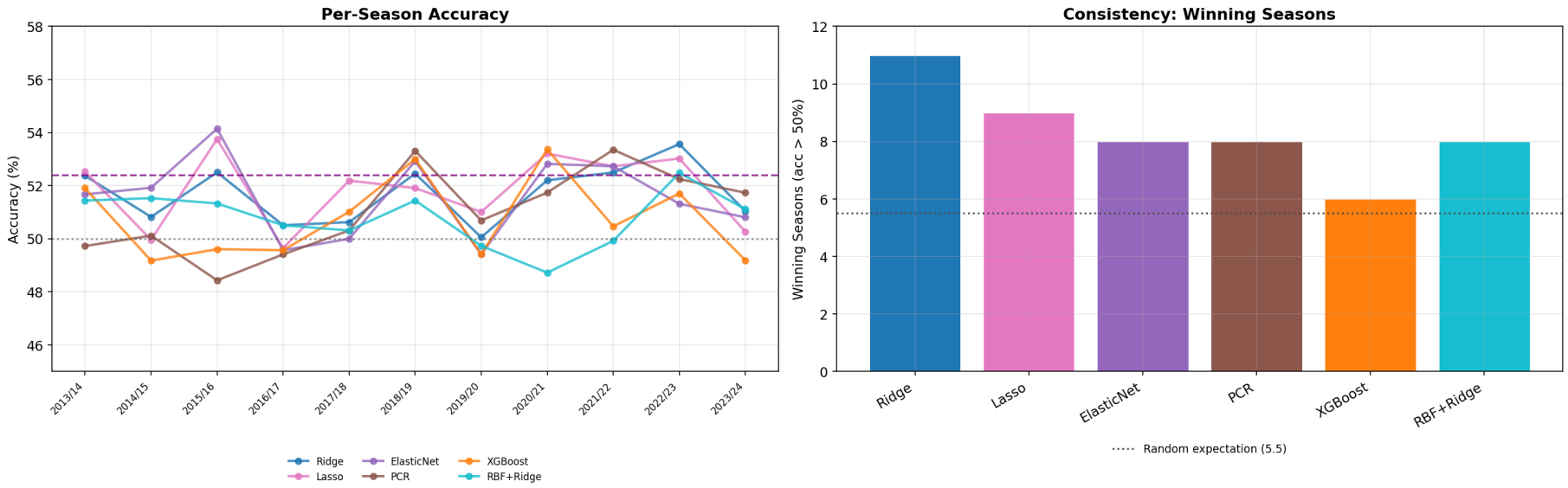
Figure 5: Model Comparison — Walk-Forward Results (2013/14 to 2023/24)



Per-Season Accuracy

- Season accuracy volatile: 48–55% range; no model consistently beats 52.4% across all years
- Lasso most consistent: above 50% in 10 of 11 seasons; linear models move together (similar signal)

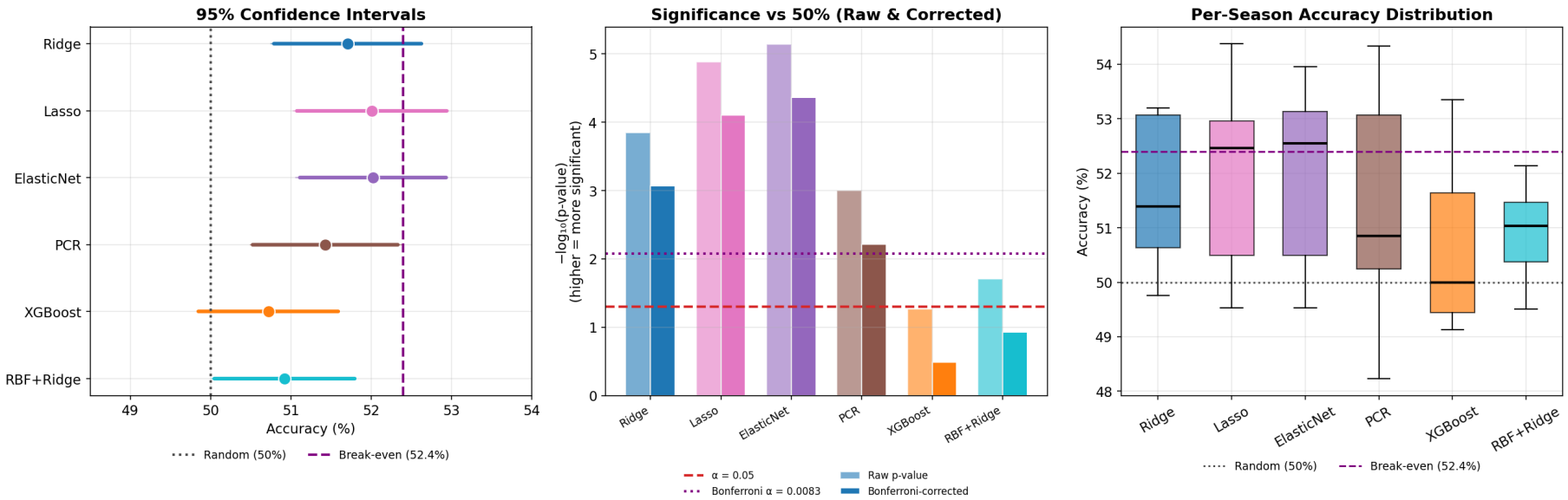
Figure 6: Per-Season Model Performance



Statistical Significance

- Wilson CIs, bootstrap, Bonferroni correction for 6 comparisons
- 4 models survive: Ridge, Lasso, ElasticNet, PCR (all $p < 0.006$ vs 50%)
- XGBoost fails significance ($p = 0.054$); every CI includes 52.4%

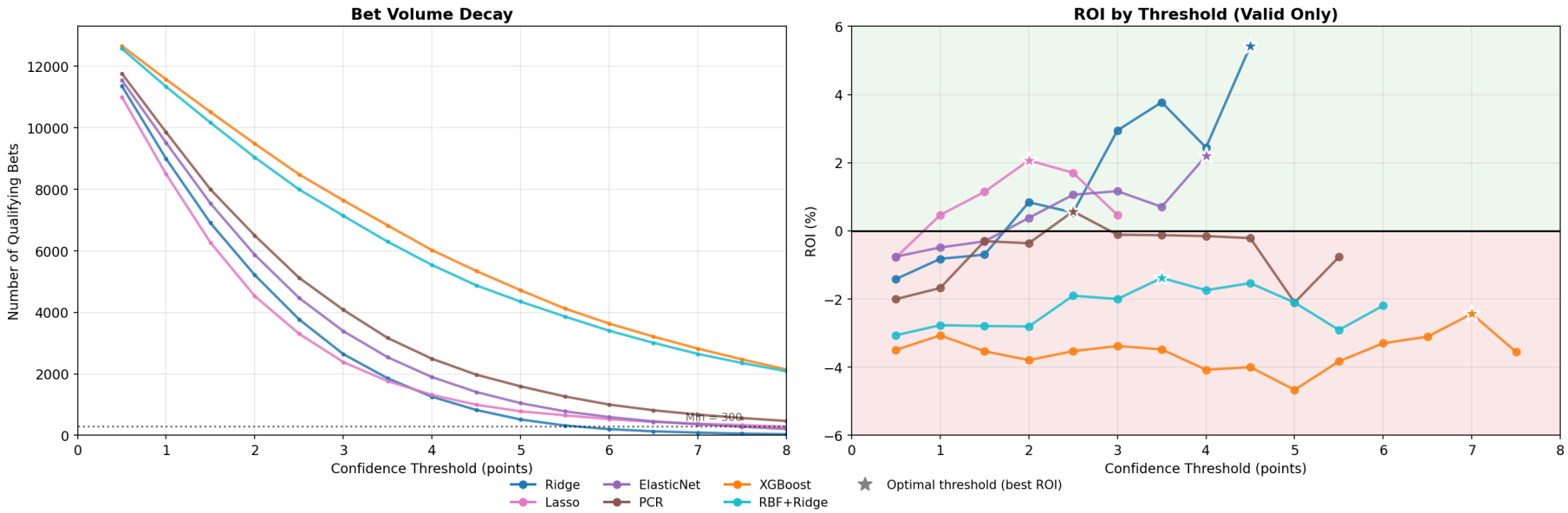
Figure 9: Statistical Validation of Model Performance



Selective Betting: Concentrating the Edge

- Bet only when model strongly disagrees with Vegas (threshold sweep 0.5–20 pts)
- 4 models achieve positive ROI; Ridge +5.4% (826 bets), Lasso +2.1% (4,529 bets)
- Linear models: conservative predictions (σ ≈ 2.5 pts) signal genuine mispricing; XGBoost: wild swings (σ ≈ 6 pts) from overfitting

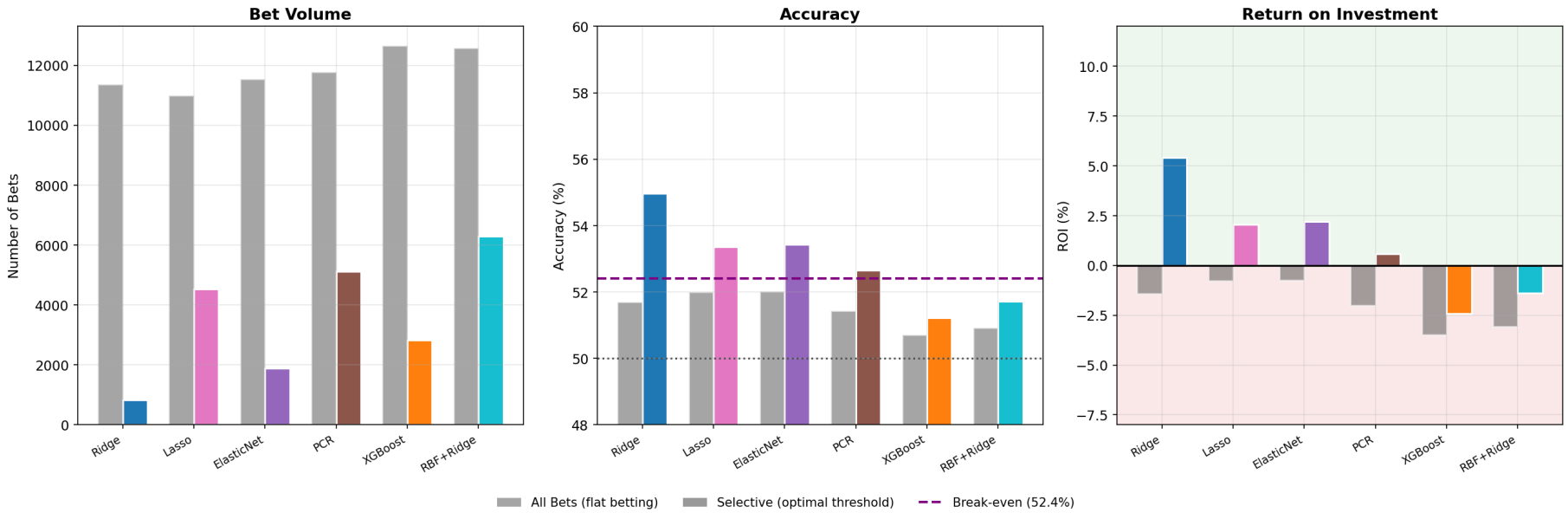
Figure 12: Threshold Overview — Volume and ROI



Base vs. Selective Performance

- Selective betting lifts 4 models above 52.4% break-even
- Not yet statistically significant above 52.4% due to smaller sample sizes

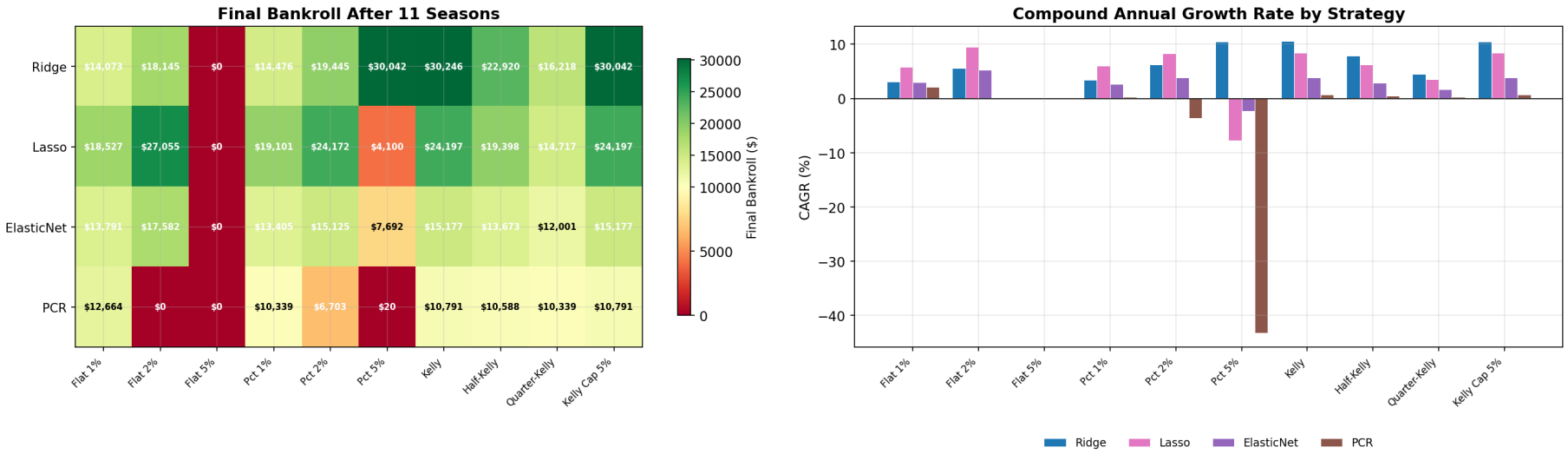
Figure 14: All Bets vs Selective Betting



Bankroll Simulation & Strategy Selection

- 10 sizing rules: flat dollar, flat %, Kelly, fractional Kelly
- Flat $500 → bankruptcy; Full Kelly → extreme drawdowns (76.6%)
- Prospective nomination: Lasso + Flat 2% ($200/bet) — lower drawdown, 5.5× more bets, simpler rule

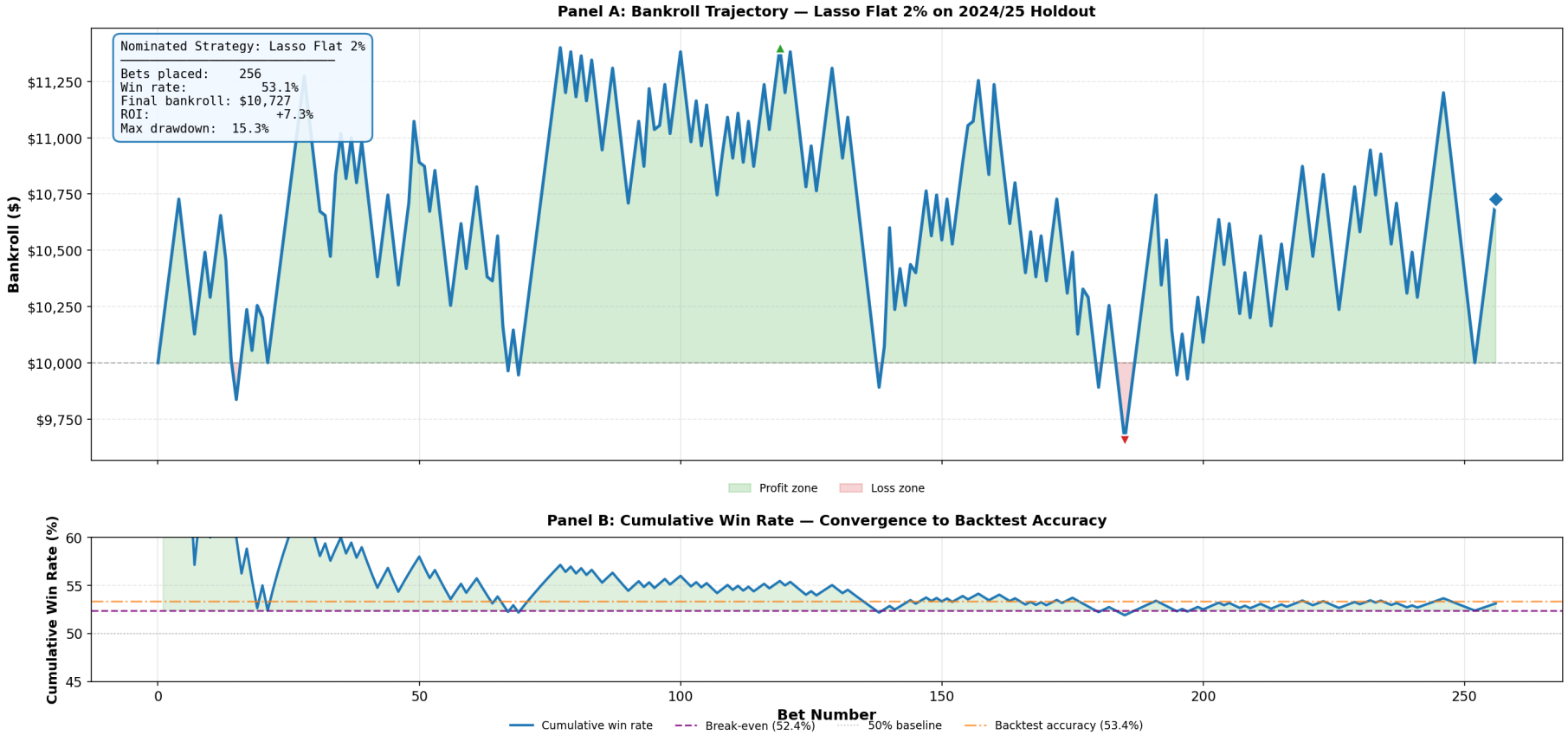
Figure 16: Backtest Bet-Sizing Comparison (11 Walk-Forward Seasons)



Holdout: 2024/25 Season Results

- Lasso: 53.1% accuracy on 256 unseen bets (backtest: 53.4%)
- $10,000 → $10,727 = +7.3% return; max drawdown 15.3%; outperforms S&P 500 (+3.7%)
- Caveat: 256 bets underpowered; 95% CI spans [47.0%, 59.1%]

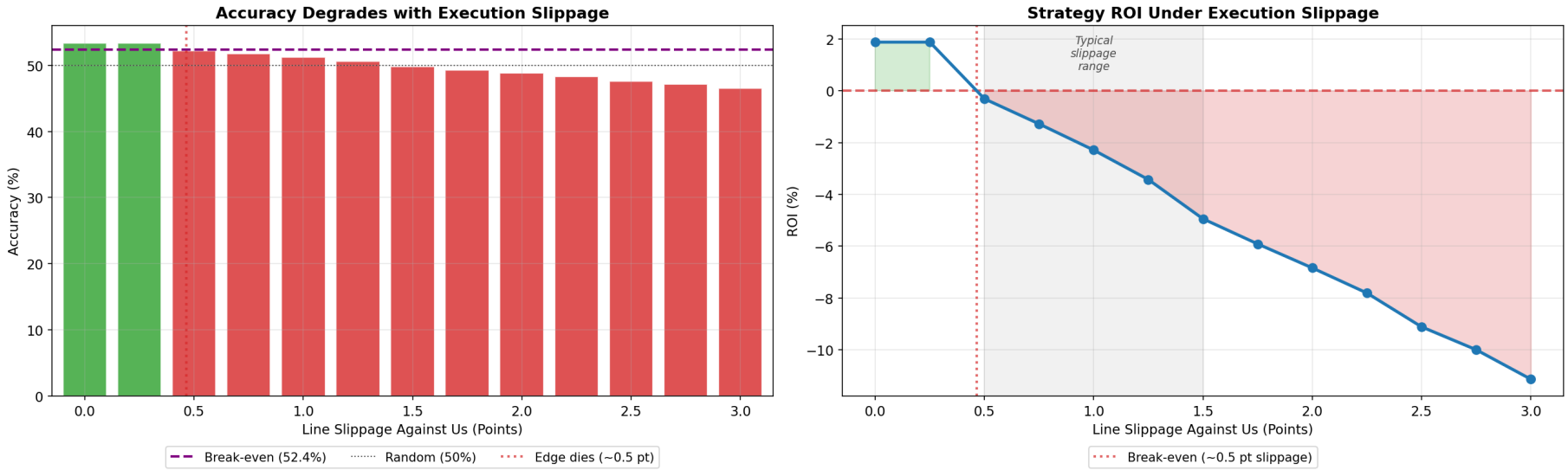
Figure 19: Lasso Deployment on 2024/25 Holdout — Prospectively Nominated Strategy



Real-World Implementation Constraints

- Slippage: break-even at ~0.5 pts; typical NBA movement 0.5–1.5 pts
- Scalability: $10K bankroll = ~$188/yr profit; ceiling ~$2.5M
- Account restrictions: Closing Line Value tracking; accounts limited in 1–3 seasons

Figure 21 — Edge Fragility: How Slippage Erodes the Lasso Advantage



Conclusions

- The edge is real: linear models at 51.7–52.0%, $p < 0.001$ after Bonferroni
- Simple models dominate: bias-variance tradeoff in a low-signal, efficient market
- Holdout confirms backtest: 53.1% accuracy, +7.3% ROI on unseen 2024/25 data
- Real-world viability is limited: slippage, betting limits, account restrictions
- Key insight: statistical significance ≠ economic significance — the defining challenge of applied ML in efficient markets

Questions?

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